Linear Systems

Theory, Fundamental Solutions,
Matrix Exponentials & Solved Exercises

Lecture Notes with Complete Solutions

Based on handwritten notes dated 12–13 March 2026

March 24, 2026

Contents

1 Linear Systems of ODEs **2**
- 1.1 Existence, Uniqueness and Solution Spaces . . . 2
- 1.2 Fundamental Solutions . . . 3
- 1.3 Variation of Parameters . . . 5
- 1.4 The Set of All Fundamental Solutions . . . 5
- 1.5 Liouville's Formula . . . 6

2 The Matrix Exponential **8**
- 2.1 Definition and Basic Properties . . . 8
- 2.2 Computing Matrix Exponentials . . . 9
- 2.3 The Inverse and Commutativity . . . 10
- 2.4 Determinant of the Matrix Exponential . . . 12
- 2.5 Jordan–Chevalley Decomposition . . . 13

3 Putzer's Method **14**
- 3.1 The Algorithm . . . 14
- 3.2 The 2×2 Case . . . 15

4 Duhamel's Formula and Commuting Systems **17**
- 4.1 $\mathcal{E}(t) = e^{tA}$ as a Fundamental Solution . . . 17
- 4.2 Duhamel's Formula . . . 17
- 4.3 Solved Exercises on Differentiability of Matrix Functions . . . 18
- 4.4 Commuting Time-Dependent Coefficient Matrices . . . 20

A Summary of Key Formulas **21**

Chapter 1

Linear Systems of ODEs

1.1 Existence, Uniqueness and Solution Spaces

Theorem 1.1.1: Existence & Uniqueness for Linear Systems

Let $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an open interval, let $A \in C(I, M_n(\mathbb{K}))$, $b \in C(I, \mathbb{K}^n)$, and let $(t_0, x_0) \in I \times \mathbb{K}^n$. Then there exists a *unique* $x \in C^1(I, \mathbb{K}^n)$ satisfying

$$\begin{cases} x'(t) = A(t)x(t) + b(t), & \forall\, t \in I, \\ x(t_0) = x_0. \end{cases}$$

Definition 1.1.1: Solution Spaces $S(I; A, b)$ and $S(I; A)$

Let $n \in \mathbb{N}$, $I \subseteq \mathbb{R}$ open, $A \in C(I, M_n(\mathbb{K}))$, $b \in C(I, \mathbb{K}^n)$. Define

$$S(I;\, A, b) \overset{\text{def}}{=} \left\{ x \in C^1(I, \mathbb{K}^n) \;\middle|\; x'(t) = A(t)x(t) + b(t),\ \forall\, t \in I \right\}.$$

For the homogeneous case we write

$$S(I;\, A) \overset{\text{def}}{=} S(I;\, A, 0).$$

Theorem 1.1.2: $S(I; A)$ is an n-dimensional $\mathbb{K}$-linear space

Let $n \in \mathbb{N}$, $I \subseteq \mathbb{R}$ open, $A \in C(I, M_n(\mathbb{K}))$. Then $S(I;\, A)$ is a $\mathbb{K}$-linear space of dimension n.

Proof

Step 1 – $S(I; A)$ is a $\mathbb{K}$-linear space. If $x, y \in S(I;\, A)$ and $\alpha, \beta \in \mathbb{K}$ then $(\alpha x + \beta y)' = \alpha x' + \beta y' = \alpha A x + \beta A y = A(\alpha x + \beta y)$, so $\alpha x + \beta y \in S(I;\, A)$. Hence $S(I;\, A)$ is a $\mathbb{K}$-linear space.

Step 2 – Constructing a basis. Fix $t_0 \in I$. Let $\{a^1, \ldots, a^n\}$ be the standard basis of $\mathbb{K}^n$. For each $i \in \{1, \ldots, n\}$ let $x^i \in C^1(I, \mathbb{K}^n)$ be the unique solution (by

Theorem 1.1.1) of

$$\begin{cases} (x^i)'(t) = A(t)x^i(t), & t \in I, \\ x^i(t_0) = a^i. \end{cases}$$

Step 3 – Linear independence. Suppose $\sum_{i=1}^{n} \alpha_i x^i = 0$ in I. Evaluating at t_0:

$$0 = \sum_{i=1}^{n} \alpha_i x^i(t_0) = \sum_{i=1}^{n} \alpha_i a^i.$$

Since $\{a^1, \dots, a^n\}$ is linearly independent, $\alpha_1 = \cdots = \alpha_n = 0$. Hence $\{x^1, \dots, x^n\}$ is linearly independent.

Step 4 – Spanning. Let $Y \in S(I;\, A)$. Write $Y(t_0) = \sum_{i=1}^{n} \beta_i a^i$ for some $\beta_i \in \mathbb{K}$. Set $Z = Y - \sum_{i=1}^{n} \beta_i x^i$. Then

$$Z'(t) = Y'(t) - \sum \beta_i (x^i)'(t) = A(t)Y(t) - \sum \beta_i A(t) x^i(t) = A(t)Z(t), \qquad Z(t_0) = 0.$$

By uniqueness (Theorem 1.1.1), $Z \equiv 0$ on I, i.e. $Y = \sum_{i=1}^{n} \beta_i x^i$. Therefore $\{x^1, \dots, x^n\}$ spans $S(I;\, A)$.

Combining Steps 3 and 4, $\{x^1, \dots, x^n\}$ is a basis of $S(I;\, A)$, so $\dim S(I;\, A) = n$. ■

1.2 Fundamental Solutions

Definition 1.2.1: Fundamental Solution Matrix

Let $n \in \mathbb{N}$, $I \subseteq \mathbb{R}$ open, $A \in C(I, M_n(\mathbb{K}))$. A matrix $F \in C^1(I, M_n(\mathbb{K}))$ is called a *fundamental solution* in I of $X'(t) = A(t)X(t)$ if $\{Fe^1, Fe^2, \dots, Fe^n\}$ forms a basis of $S(I;\, A)$, where $\{e^1, \dots, e^n\}$ is the standard (orthonormal) basis of $\mathbb{K}^n$. The collection of all fundamental solutions is denoted

$$\mathcal{F}(I, A) = \left\{ F \in C^1(I, M_n(\mathbb{K})) \;\middle|\; F \text{ is a fundamental solution in } I \text{ of } X'(t) = A(t)X(t) \right\}.$$

Lemma 1.2.1: Properties of $F \in \mathcal{F}(I, A)$

Let $n \in \mathbb{N}$, $I \subseteq \mathbb{R}$ open, $A \in C(I, M_n(\mathbb{K}))$, $F \in \mathcal{F}(I, A)$. Then

(i) $F'(t) = A(t)F(t)$ for all $t \in I$.

(ii) $S(I;\, A) = \{Fc \mid c \in \mathbb{K}^n\}$.

(iii) $\operatorname{rank} F(t) = n$ for all $t \in I$.

Proof

(i) For each $j \in \{1, \dots, n\}$, since $Fe^j \in S(I;\, A)$:

$$F'(t)e^j = (F(t)e^j)' = A(t)F(t)e^j \implies (F'(t) - A(t)F(t))e^j = 0.$$

Since this holds for all basis vectors e^j, we conclude $F'(t) = A(t)F(t)$.

(ii) See Exercise 1.2.1 below (solved).

(iii) Suppose for contradiction that $\operatorname{rank} F(t_0) < n$ for some $t_0 \in I$. Then there exists $a \in \mathbb{K}^n \backslash \{0\}$ with $F(t_0)a = 0$. By part (i), $Fa \in S(I;\, A)$ (the map $t \mapsto F(t)a$ satisfies the homogeneous equation). By uniqueness, $F(t)a = 0$ for all $t \in I$, so $\sum_j \alpha_j Fe^j = 0$ in I (where $a = \sum_j \alpha_j e^j$, not all α_j zero). This contradicts $\{Fe^j\}$ being linearly independent. ■

Exercise 1.2.1: Proof of $S(I; A) = \{Fc \mid c \in \mathbb{K}^n\}$

Let $F \in \mathcal{F}(I, A)$. Show that $S(I;\, A) = \{Fc \mid c \in \mathbb{K}^n\}$.

Solution

We show the two inclusions.
($\supseteq$) Let $c \in \mathbb{K}^n$ and set $x(t) = F(t)c$. By Lemma 1.2.1(i), $x'(t) = F'(t)c = A(t)F(t)c = A(t)x(t)$. Hence $x \in S(I;\, A)$, so $\{Fc \mid c \in \mathbb{K}^n\} \subseteq S(I;\, A)$.
($\subseteq$) Let $Y \in S(I;\, A)$. Since $\{Fe^1, \dots, Fe^n\}$ is a basis of $S(I;\, A)$ by definition of fundamental solution, there exist $\beta_1, \dots, \beta_n \in \mathbb{K}$ such that

$$Y = \sum_{i=1}^{n} \beta_i Fe^i = F\left(\sum_{i=1}^{n} \beta_i e^i\right) = Fc,$$

where $c = \sum_{i=1}^{n} \beta_i e^i \in \mathbb{K}^n$. Thus $Y \in \{Fc \mid c \in \mathbb{K}^n\}$.
Combining both inclusions, $S(I;\, A) = \{Fc \mid c \in \mathbb{K}^n\}$.

Lemma 1.2.2: Characterisation of Fundamental Solutions

Let $F \in C^1(I, M_n(\mathbb{K}))$. Then $F \in \mathcal{F}(I, A)$ if and only if

(i) $F'(t) = A(t)F(t)$ for all $t \in I$, and

(ii) $\operatorname{rank} F(t_0) = n$ for some $t_0 \in I$.

Proof

($\Rightarrow$) Follows immediately from Lemma 1.2.1(i) and (iii).

($\Leftarrow$) Assume (i) and (ii). For each j, the column Fe^j satisfies $(Fe^j)' = F'e^j = AFe^j$, so $Fe^j \in S(I;\, A)$.
It remains to show $\{Fe^1, \dots, Fe^n\}$ is linearly independent. Suppose $\sum_j c_j Fe^j = 0$ in I. Evaluating at t_0: $F(t_0)\left(\sum_j c_j e^j\right) = 0$. Since $\operatorname{rank} F(t_0) = n$, the matrix $F(t_0)$

is invertible, so $\sum_j c_j e^j = 0$, whence $c_j = 0$ for all j. Hence $\{Fe^1, \ldots, Fe^n\}$ is linearly independent, giving $\dim S(I;\, A) = n$. By Theorem 1.1.2 this means they form a basis, so $F \in \mathcal{F}(I, A)$. ■

1.3 Variation of Parameters

Theorem 1.3.1: General Solution Formula

Let $n \in \mathbb{N}$, $I \subseteq \mathbb{R}$ open, $A \in C(I, M_n(\mathbb{K}))$, $b \in C(I, \mathbb{K}^n)$, $(t_0, x_0) \in I \times \mathbb{K}^n$, and $F \in \mathcal{F}(I, A)$. Then the unique solution of

$$\begin{cases} x'(t) = A(t)x(t) + b(t), \\ x(t_0) = x_0 \end{cases}$$

is given by

$$\boxed{x(t) = F(t)F(t_0)^{-1}x_0 + \int_{t_0}^{t} F(t)F(s)^{-1}b(s)\,ds, \qquad \forall\, t \in I.}$$

Proof

Note that $F(t_0)^{-1}$ exists by Lemma 1.2.1(iii). Differentiate x:

$$\begin{aligned} x'(t) &= F'(t)F(t_0)^{-1}x_0 + F(t)F(t)^{-1}b(t) + \int_{t_0}^{t} F'(t)F(s)^{-1}b(s)\,ds \\ &= A(t)F(t)F(t_0)^{-1}x_0 + b(t) + \int_{t_0}^{t} A(t)F(t)F(s)^{-1}b(s)\,ds \\ &= A(t)\left[F(t)F(t_0)^{-1}x_0 + \int_{t_0}^{t} F(t)F(s)^{-1}b(s)\,ds\right] + b(t) \\ &= A(t)x(t) + b(t). \end{aligned}$$

Also $x(t_0) = F(t_0)F(t_0)^{-1}x_0 = x_0$. Uniqueness follows from Theorem 1.1.1. ■

1.4 The Set of All Fundamental Solutions

Exercise 1.4.1: $\mathcal{F}(I, A) = \{FC \mid C \in \mathrm{GL}_n(\mathbb{K})\}$

Let $F \in \mathcal{F}(I, A)$. Show that

$$\mathcal{F}(I, A) = \{FC \mid C \in \mathrm{GL}_n(\mathbb{K})\}.$$

Solution

($\supseteq$) Let $C \in \mathrm{GL}_n(\mathbb{K})$ and $G = FC$. Then:

- $G'(t) = F'(t)C = A(t)F(t)C = A(t)G(t)$.
- $\operatorname{rank} G(t_0) = \operatorname{rank}(F(t_0)C) = n$ since both $F(t_0)$ and C are invertible.

By Lemma 1.2.2, $G \in \mathcal{F}(I, A)$.
($\subseteq$) Let $G \in \mathcal{F}(I, A)$. By Lemma 1.2.1(ii) applied to both F and G:

$$S(I;\, A) = \{Fc \mid c \in \mathbb{K}^n\} = \{Gc \mid c \in \mathbb{K}^n\}.$$

For each j, $Ge^j \in S(I;\, A)$, so $Ge^j = Fc^j$ for some $c^j \in \mathbb{K}^n$. Define $C \in M_n(\mathbb{K})$ by $Ce^j = c^j$, i.e. the j-th column of C is c^j. Then $G = FC$.
It remains to show $C \in \mathrm{GL}_n(\mathbb{K})$. Since $\operatorname{rank} G(t_0) = n$ and $G(t_0) = F(t_0)C$, and $F(t_0)$ is invertible, we get $\operatorname{rank} C = n$, so $C \in \mathrm{GL}_n(\mathbb{K})$.

1.5 Liouville's Formula

Lemma 1.5.1: Multilinearity of Determinant

Let $\{a_1, \ldots, a_n\}$ be a basis of $\mathbb{K}^n$ and let $T : \mathbb{K}^n \to \mathbb{K}^n$ be $\mathbb{K}$-linear. Then

$$\sum_{j=1}^{n} \det(a_1, \ldots, Ta_j, \ldots, a_n) = (\operatorname{tr} T)\, \det(a_1, \ldots, a_n).$$

Proof

Write $Ta_j = \sum_{i=1}^{n} t_{ij} a_i$. Then by multilinearity and alternating property of det:

$$\begin{aligned}
\sum_j \det(\ldots, Ta_j, \ldots) &= \sum_j \det\Big(\ldots, \sum_i t_{ij} a_i, \ldots\Big) \\
&= \sum_j \sum_i t_{ij} \det(\ldots, a_i, \ldots) = \sum_j t_{jj} \det(a_1, \ldots, a_n) \\
&= \Big(\sum_j t_{jj}\Big) \det(a_1, \ldots, a_n) = (\operatorname{tr} T) \det(a_1, \ldots, a_n).
\end{aligned}$$

(Terms with $i \neq j$ vanish because two columns are equal.) ■

Theorem 1.5.1: Liouville's Formula

Let $n \in \mathbb{N}$, $I \subseteq \mathbb{R}$ open, $A \in C(I, M_n(\mathbb{K}))$, $F \in \mathcal{F}(I, A)$, and $t_0 \in I$. Then

$$\boxed{\det F(t) = \det F(t_0) \cdot \exp\left(\int_{t_0}^{t} \operatorname{tr}(A(s))\, ds\right), \qquad \forall\, t \in I.}$$

Proof

Write $F(t) = [F(t)e^1, \dots, F(t)e^n]$. Differentiating the determinant (column by column):

$$\begin{aligned}\frac{d}{dt}\det F(t) &= \sum_{j=1}^{n} \det\big[F(t)e^1, \dots, (F(t)e^j)', \dots, F(t)e^n\big] \\ &= \sum_{j=1}^{n} \det\big[F(t)e^1, \dots, A(t)F(t)e^j, \dots, F(t)e^n\big] \\ &= \operatorname{tr}(A(t)) \cdot \det F(t),\end{aligned}$$

where the last step uses Lemma 1.5.1 with $a_k = F(t)e^k$ and $T = A(t)$.
This is a scalar linear ODE: $\frac{d}{dt}f = (\operatorname{tr} A(t))f$, with solution $f(t) = f(t_0)\exp\big(\int_{t_0}^{t} \operatorname{tr} A(s)\,ds\big)$. ■

Chapter 2

The Matrix Exponential

2.1 Definition and Basic Properties

Definition 2.1.1: Matrix Exponential

Let $n \in \mathbb{N}$ and $A \in M_n(\mathbb{K})$. The *exponential* of A is defined by

$$e^A \stackrel{\text{def}}{=} \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

Remark 2.1.1: Well-definedness

The series converges absolutely in $M_n(\mathbb{K})$ (with any sub-multiplicative norm) since

$$\sum_{k=0}^{\infty} \left\| \frac{A^k}{k!} \right\| \leq \sum_{k=0}^{\infty} \frac{\|A\|^k}{k!} = e^{\|A\|} < \infty.$$

In particular, $\|e^A\| \leq e^{\|A\|}$.

Lemma 2.1.1: Derivative of $t \mapsto e^{tA}$

Let $\mathcal{E} : \mathbb{R} \to M_n(\mathbb{K})$ be defined by $\mathcal{E}(t) = e^{tA}$. Then

$$\mathcal{E}'(t) = A\mathcal{E}(t) = \mathcal{E}(t)A, \qquad \forall\, t \in \mathbb{R}.$$

Proof

Differentiating the power series term by term (justified by uniform convergence on compact sets):

$$\mathcal{E}'(t) = \sum_{k=1}^{\infty} \frac{kt^{k-1}A^k}{k!} = \sum_{k=1}^{\infty} \frac{t^{k-1}A^k}{(k-1)!} = A \sum_{j=0}^{\infty} \frac{t^j A^j}{j!} = A\mathcal{E}(t) = \mathcal{E}(t)A.$$

■

Theorem 2.1.1: Unique Solution via Matrix Exponential

Let $A \in M_n(\mathbb{K})$, $x_0 \in \mathbb{K}^n$, and $I \subseteq \mathbb{R}$ open with $0 \in I$. Then

$$x(t) = e^{tA}x_0, \qquad t \in I,$$

is the unique solution in I of

$$\begin{cases} x'(t) = Ax(t), \\ x(0) = x_0. \end{cases}$$

Proof

Let x be any solution. Define $F(t) = e^{-tA}x(t)$. Then

$$F'(t) = e^{-tA}(-A)x(t) + e^{-tA}x'(t) = e^{-tA}(-Ax(t) + Ax(t)) = 0, \qquad \forall\, t \in I.$$

Hence F is constant, and $F(0) = e^0 x_0 = x_0$. Therefore $e^{-tA}x(t) = x_0$, giving $x(t) = e^{tA}x_0$. ■

2.2 Computing Matrix Exponentials

Example 2.2.1: Diagonal Matrices

Let $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Then $A^k = \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k)$, so

$$e^A = \operatorname{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n}).$$

Example 2.2.2: Nilpotent Matrices

Let $A \in M_n(\mathbb{K})$ be nilpotent, i.e. $A^n = 0$. Then

$$e^A = \sum_{k=0}^{n-1} \frac{A^k}{k!}.$$

In particular, for $n = 2$:

$$\boxed{e^A = I + A.}$$

Example 2.2.3: Conformal (Rotation–Scaling) Matrices

Let $a, b \in \mathbb{R}$ and $A(a,b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. Write $\lambda = a + ib \in \mathbb{C}$. One verifies by induction that

$$A(a,b)^k = \begin{pmatrix} \operatorname{Re}\lambda^k & -\operatorname{Im}\lambda^k \\ \operatorname{Im}\lambda^k & \operatorname{Re}\lambda^k \end{pmatrix}.$$

Summing the series:

$$e^{A(a,b)} = \begin{pmatrix} \operatorname{Re} e^{\lambda} & -\operatorname{Im} e^{\lambda} \\ \operatorname{Im} e^{\lambda} & \operatorname{Re} e^{\lambda} \end{pmatrix} = e^{a} \begin{pmatrix} \cos b & -\sin b \\ \sin b & \cos b \end{pmatrix}.$$

Non-injectivity: $e^{A(a,b+2k\pi)} = e^{A(a,b)}$ for all $k \in \mathbb{Z}$, so $\exp : M_2(\mathbb{R}) \to M_2(\mathbb{R})$ is *not* injective.

2.3 The Inverse and Commutativity

Lemma 2.3.1: $e^{tA}e^{-tA} = I$

For all $A \in M_n(\mathbb{K})$ and $t \in \mathbb{R}$, $e^{tA}e^{-tA} = I$.

Proof

Let $F(t) = e^{tA}e^{-tA}$. Then

$$F'(t) = e^{tA}Ae^{-tA} + e^{tA}(-A)e^{-tA} = 0,$$

so F is constant. At $t = 0$, $F(0) = I$. Thus $e^{tA}e^{-tA} = I$ for all t. In particular $(e^{A})^{-1} = e^{-A}$, so $e^{A} \in \mathrm{GL}_n(\mathbb{K})$. ■

Theorem 2.3.1: Exponential of a Sum (Commuting Case)

Let $A, B \in M_n(\mathbb{K})$ satisfy $AB = BA$. Then

$$e^{A+B} = e^{A}e^{B}.$$

Remark 2.3.1: The Theorem Fails without Commutativity

Consider $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Note $AB \neq BA$. One can compute

$$e^{A} = \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix}, \quad e^{B} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad e^{A+B} = \begin{pmatrix} e & e-1 \\ 0 & 1 \end{pmatrix},$$

and verify $e^{A+B} \notin \{e^{A}e^{B}, e^{B}e^{A}\}$ (see solved exercise below).

Proof

[Proof of Theorem 2.3.1] Define $F : \mathbb{R} \to M_n(\mathbb{K})$ by $F(t) = e^{t(A+B)} - e^{tA}e^{tB}$. Since $AB = BA$ we have $Be^{tA} = e^{tA}B$ (which follows by comparing power series), so:

$$\begin{aligned} F'(t) &= (A+B)e^{t(A+B)} - Ae^{tA}e^{tB} - e^{tA}Be^{tB} \\ &= (A+B)e^{t(A+B)} - (A+B)e^{tA}e^{tB} = (A+B)F(t). \end{aligned}$$

Also $F(0) = 0$. By uniqueness (linear IVP), $F(t) = 0$ for all t, giving $e^{A+B} = e^A e^B$. ■

Exercise 2.3.1: Counterexample when $AB \neq BA$

Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

(i) Compute e^A, e^B, and e^{A+B}.

(ii) Show that $e^{A+B} \notin \{e^A e^B, e^B e^A\}$.

Solution

(i)
Computing e^A: A is diagonal with eigenvalues 1 and 0, so by Example 2.2.1:

$$e^A = \operatorname{diag}(e^1, e^0) = \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix}.$$

Computing e^B: Note $B^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0$, so B is nilpotent with $B^2 = 0$. By Example 2.2.2:

$$e^B = I + B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Computing e^{A+B}: $A + B = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$. Compute $(A+B)^k$: one checks $(A+B)^k = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ for all $k \geq 1$. Hence

$$e^{A+B} = I + \sum_{k=1}^{\infty} \frac{(A+B)^k}{k!} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} \sum_{k=1}^{\infty} \frac{1}{k!} & \sum_{k=1}^{\infty} \frac{1}{k!} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} e & e-1 \\ 0 & 1 \end{pmatrix}.$$

(ii) Compute the two products:

$$e^A e^B = \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e & e \\ 0 & 1 \end{pmatrix},$$

$$e^B e^A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e & 1 \\ 0 & 1 \end{pmatrix}.$$

Comparing:

$$e^{A+B} = \begin{pmatrix} e & e-1 \\ 0 & 1 \end{pmatrix}, \quad e^A e^B = \begin{pmatrix} e & e \\ 0 & 1 \end{pmatrix}, \quad e^B e^A = \begin{pmatrix} e & 1 \\ 0 & 1 \end{pmatrix}.$$

Since $e - 1 \neq e$ and $e - 1 \neq 1$ (as $e \neq 1$ and $e \neq 2$), we conclude $e^{A+B} \notin \{e^A e^B, e^B e^A\}$.

2.4 Determinant of the Matrix Exponential

Theorem 2.4.1: $\det e^A = e^{\operatorname{tr} A}$

For all $A \in M_n(\mathbb{K})$,

$$\boxed{\det e^A = e^{\operatorname{tr} A}.}$$

Lemma 2.4.1: Similarity and Exponential

Let $A \in M_n(\mathbb{K})$ and $P \in \mathrm{GL}_n(\mathbb{K})$. Then

$$e^{PAP^{-1}} = Pe^A P^{-1}.$$

Proof

Since $(PAP^{-1})^k = PA^kP^{-1}$ for all $k \in \mathbb{N}$,

$$e^{PAP^{-1}} = \sum_{k=0}^{\infty} \frac{(PAP^{-1})^k}{k!} = P\left(\sum_{k=0}^{\infty} \frac{A^k}{k!}\right) P^{-1} = Pe^A P^{-1}.$$

■

Proof

[Proof of Theorem 2.4.1] We consider three cases.

Case 1: A is diagonalisable over $\mathbb{C}$. Find $P \in \mathrm{GL}_n(\mathbb{C})$ and $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ with $A = P \operatorname{diag}(\lambda_1, \ldots, \lambda_n)P^{-1}$. By Lemma 2.4.1:

$$e^A = P \operatorname{diag}(e^{\lambda_1}, \ldots, e^{\lambda_n})P^{-1},$$

$$\det e^A = \det P \cdot \prod_{i=1}^{n} e^{\lambda_i} \cdot (\det P)^{-1} = e^{\lambda_1 + \cdots + \lambda_n} = e^{\operatorname{tr} A}.$$

Case 2: A is nilpotent. Since $A^n = 0$, we have $e^A = I + A + \cdots + \frac{A^{n-1}}{(n-1)!}$, so $(e^A - I)$ is nilpotent (as a polynomial in A with zero constant term). Hence $(e^A - I)^n = 0$, meaning $(\lambda - 1)^n$ is an annihilating polynomial of e^A, so $\sigma(e^A) = \{1\}$. Therefore $\det e^A = 1 = e^0 = e^{\operatorname{tr} A}$ (since $\operatorname{tr} A = 0$ for nilpotent A).

Case 3: General case. By the Jordan–Chevalley decomposition (Theorem 2.5.1), write $A = D + N$ with D diagonalisable, N nilpotent, $DN = ND$. Note that $\operatorname{tr} A = \operatorname{tr} D$. By Theorem 2.3.1:

$$\det e^A = \det(e^D e^N) = \det e^D \cdot \det e^N = e^{\operatorname{tr} D} \cdot 1 = e^{\operatorname{tr} A}.$$

■

2.5 Jordan–Chevalley Decomposition

Theorem 2.5.1: Jordan–Chevalley Decomposition

Let $A \in M_n(\mathbb{K})$. Then there exist unique $D, N \in M_n(\mathbb{K})$ such that

(i) $A = D + N$,

(ii) D is diagonalisable over $\mathbb{C}$,

(iii) N is nilpotent,

(iv) $DN = ND$.

Furthermore, both D and N are polynomials in A.

Chapter 3

Putzer's Method

3.1 The Algorithm

Theorem 3.1.1: Putzer's Method

Let $A \in M_n(\mathbb{K})$ and let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A (counted with multiplicity, in any order). Define

$$P_k(A) = \begin{cases} I & k = 0, \\ \prod_{i=1}^{k} (A - \lambda_i I) & 1 \leq k \leq n, \end{cases}$$

and let $r_k : \mathbb{R} \to \mathbb{C}$ be defined recursively by

$$r_1'(t) = \lambda_1 r_1(t), \quad r_1(0) = 1,$$

$$r_{k+1}'(t) = \lambda_{k+1} r_{k+1}(t) + r_k(t), \quad r_{k+1}(0) = 0, \quad k = 1, \ldots, n-1.$$

Then

$$\boxed{e^{tA} = \sum_{k=0}^{n-1} r_{k+1}(t)\, P_k(A), \qquad \forall\, t \in \mathbb{R}.}$$

(By Cayley–Hamilton, $P_n(A) = 0$, which is used in the proof.)

Proof

Define $F(t) = \sum_{k=0}^{n-1} r_{k+1}(t) P_k(A)$. We show $F'(t) = AF(t)$ and $F(0) = I$.
Initial condition: $F(0) = r_1(0) P_0(A) = 1 \cdot I = I$.

Differential equation: Setting $r_0 \equiv 0$:

$$\begin{aligned} F'(t) &= \sum_{k=0}^{n-1} r'_{k+1}(t)P_k(A) = \sum_{k=0}^{n-1}(\lambda_{k+1}r_{k+1}(t) + r_k(t))P_k(A) \\ &= \sum_{k=0}^{n-1} \lambda_{k+1}r_{k+1}(t)P_k(A) + \sum_{k=0}^{n-2} r_{k+1}(t)P_{k+1}(A). \end{aligned}$$

Using $P_{k+1}(A) = (A - \lambda_{k+1}I)P_k(A)$:

$$\begin{aligned} F'(t) - \lambda_n F(t) &= \sum_{k=0}^{n-2}(\lambda_{k+1} - \lambda_n)r_{k+1}P_k(A) + \sum_{k=0}^{n-2} r_{k+1}(A - \lambda_{k+1}I)P_k(A) \\ &= \sum_{k=0}^{n-2} r_{k+1}\big[(\lambda_{k+1} - \lambda_n) + (A - \lambda_{k+1}I)\big]P_k(A) \\ &= \sum_{k=0}^{n-2} r_{k+1}(A - \lambda_n I)P_k(A) = (A - \lambda_n I)\big(F(t) - r_n(t)P_{n-1}(A)\big) \\ &= (A - \lambda_n I)F(t) - r_n(t)P_n(A) = (A - \lambda_n I)F(t), \end{aligned}$$

where we used $P_n(A) = 0$ (Cayley–Hamilton). Hence $F'(t) = AF(t)$.
Since F satisfies the same IVP as e^{tA}, uniqueness gives $F(t) = e^{tA}$. ■

3.2 The 2×2 Case

Example 3.2.1: Putzer for $n = 2$

Let $A \in M_2(\mathbb{R})$ with eigenvalues λ, μ ($\lambda \neq \mu$). Then $P_0(A) = I$, $P_1(A) = A - \lambda I$, and:

$$r_1(t) = e^{\lambda t}, \qquad r'_2(t) - \mu r_2(t) = e^{\lambda t}, \quad r_2(0) = 0.$$

Solve the r_2 equation (integrating factor $e^{-\mu t}$):

$$\frac{d}{dt}(e^{-\mu t}r_2(t)) = e^{(\lambda-\mu)t} \implies e^{-\mu t}r_2(t) = \frac{e^{(\lambda-\mu)t} - 1}{\lambda - \mu} \implies r_2(t) = \frac{e^{\lambda t} - e^{\mu t}}{\lambda - \mu}.$$

Therefore

$$\boxed{e^{tA} = e^{\lambda t}I + \frac{e^{\lambda t} - e^{\mu t}}{\lambda - \mu}(A - \lambda I), \qquad \lambda \neq \mu.}$$

Exercise 3.2.1: Putzer Formula for $\lambda = \mu$ (repeated eigenvalue)

Find the formula for e^{tA} when $A \in M_2(\mathbb{R})$ has a repeated eigenvalue $\lambda = \mu$.

Solution

In this case:

$$r_1(t) = e^{\lambda t}, \qquad r_2'(t) - \lambda r_2(t) = e^{\lambda t}, \quad r_2(0) = 0.$$

Multiplying both sides by $e^{-\lambda t}$:

$$\frac{d}{dt}(e^{-\lambda t} r_2(t)) = 1 \implies e^{-\lambda t} r_2(t) = t \implies r_2(t) = te^{\lambda t}.$$

Hence, with $P_0(A) = I$ and $P_1(A) = A - \lambda I$:

$$\boxed{e^{tA} = e^{\lambda t} I + te^{\lambda t}(A - \lambda I) = e^{\lambda t}\big(I + t(A - \lambda I)\big).}$$

Consistency check: At $t = 0$, $e^0 = I$. ✓
Differentiating at $t = 0$: $(e^{tA})'|_{t=0} = \lambda I + (A - \lambda I) = A$. ✓

Chapter 4

Duhamel's Formula and Commuting Systems

4.1 $\mathcal{E}(t) = e^{tA}$ as a Fundamental Solution

Theorem 4.1.1: $e^{tA} \in \mathcal{F}(I, A)$ (constant A)

Let $A \in M_n(\mathbb{K})$, $I \subseteq \mathbb{R}$ open, and $\mathcal{E} : I \to M_n(\mathbb{K})$, $\mathcal{E}(t) = e^{tA}$. Then $\mathcal{E} \in \mathcal{F}(I, A)$.

Proof

We verify the two conditions of Lemma 1.2.2.
(i) By Lemma 2.1.1, $\mathcal{E}'(t) = Ae^{tA} = A\mathcal{E}(t)$.
(ii) $\mathcal{E}(0) = e^0 = I$, which has rank n.
Hence $\mathcal{E} \in \mathcal{F}(I, A)$. ■

4.2 Duhamel's Formula

Theorem 4.2.1: Duhamel's Formula

Let $n \in \mathbb{N}$, $I \subseteq \mathbb{R}$ open, $A \in M_n(\mathbb{K})$, $b \in C(I, \mathbb{K}^n)$, $(t_0, x_0) \in I \times \mathbb{K}^n$. Then

$$x(t) = e^{(t-t_0)A}x_0 + \int_{t_0}^{t} e^{(t-s)A}b(s)\,ds, \qquad t \in I,$$

is the unique solution of

$$\begin{cases} x'(t) = Ax(t) + b(t), & \forall\, t \in I, \\ x(t_0) = x_0. \end{cases}$$

Proof

Take $F(t) = e^{tA} \in \mathcal{F}(I, A)$ (Theorem 4.1.1). Then $F(t)F(t_0)^{-1} = e^{tA}e^{-t_0A} = e^{(t-t_0)A}$ and $F(t)F(s)^{-1} = e^{(t-s)A}$. The result follows directly from Theorem 1.3.1. ■

4.3 Solved Exercises on Differentiability of Matrix Functions

Exercise 4.3.1: Derivative of $X(t)^m$

Let $I \subseteq \mathbb{R}$ open, $X : I \to M_n(\mathbb{K})$ differentiable at $s \in I$, and $m \in \mathbb{N}$. Define $\varphi : I \to M_n(\mathbb{K})$ by $\varphi(t) = X(t)^m$. Show that φ is differentiable at s and

$$\varphi'(s) = \sum_{k=1}^{m} X(s)^{k-1}X'(s)X(s)^{m-k}.$$

Solution

We proceed by induction on m.
Base case $m = 1$**:** $\varphi(t) = X(t)$, so $\varphi'(s) = X'(s) = \sum_{k=1}^{1} X(s)^0X'(s)X(s)^0 = X'(s)$. ✓
Inductive step: Assume the formula holds for $m-1$. Then $\varphi(t) = X(t)^{m-1} \cdot X(t)$, and by the product rule for matrix-valued functions:

$$\varphi'(s) = \left(X(s)^{m-1}\right)'X(s) + X(s)^{m-1}X'(s).$$

By the inductive hypothesis:

$$\left(X(s)^{m-1}\right)' = \sum_{k=1}^{m-1} X(s)^{k-1}X'(s)X(s)^{(m-1)-k}.$$

Therefore:

$$\begin{aligned}
\varphi'(s) &= \sum_{k=1}^{m-1} X(s)^{k-1}X'(s)X(s)^{m-1-k} \cdot X(s) + X(s)^{m-1}X'(s) \\
&= \sum_{k=1}^{m-1} X(s)^{k-1}X'(s)X(s)^{m-k} + X(s)^{m-1}X'(s)X(s)^0 \\
&= \sum_{k=1}^{m} X(s)^{k-1}X'(s)X(s)^{m-k},
\end{aligned}$$

which completes the induction.

Exercise 4.3.2: Derivative of $e^{X(t)}$

Let $f(t) = e^{X(t)}$ where $X : I \to M_n(\mathbb{K})$ is differentiable at $s \in I$. Show that f is differentiable at s and

$$f'(s) = \int_0^1 e^{\lambda X(s)} X'(s) e^{(1-\lambda)X(s)}\, d\lambda.$$

In particular, if $X(s)X'(s) = X'(s)X(s)$, then $f'(s) = e^{X(s)}X'(s) = X'(s)e^{X(s)}$.

Solution

General formula. Consider the function $G : [0,1] \to M_n(\mathbb{K})$ defined by $G(\lambda) = e^{\lambda X(s)} X'(s) e^{(1-\lambda)X(s)}$. We want to show $f'(s) = \int_0^1 G(\lambda)\, d\lambda$.
Define $H(\lambda) = e^{\lambda X(s)} f(s) e^{-(1-\lambda)X(s)}$... Instead, we use the following standard approach:
Let $h \in \mathbb{R}$ be small and write $X(s+h) = X(s) + hX'(s) + o(h)$. We wish to differentiate $e^{X(s)+hX'(s)}$ with respect to h at $h = 0$.
Consider the auxiliary function $\Phi : [0,1] \to M_n(\mathbb{K})$ defined by

$$\Phi(\lambda) = e^{X(s)+\lambda h X'(s)},$$

so $\Phi(0) = e^{X(s)}$ and $\Phi(1) = e^{X(s)+hX'(s)}$. Differentiating with respect to λ (the matrices $X(s) + \lambda h X'(s)$ and $hX'(s)$ do *not* generally commute):

$$\Phi'(\lambda) = \int_0^1 e^{\mu(X(s)+\lambda h X'(s))} (hX'(s)) e^{(1-\mu)(X(s)+\lambda h X'(s))}\, d\mu.$$

Then

$$\Phi(1) - \Phi(0) = \int_0^1 \Phi'(\lambda)\, d\lambda = \int_0^1 \int_0^1 e^{\mu X(s) + O(h)} (hX'(s)) e^{(1-\mu)X(s)+O(h)}\, d\mu\, d\lambda.$$

Dividing by h and taking $h \to 0$:

$$f'(s) = \lim_{h\to 0} \frac{e^{X(s+h)} - e^{X(s)}}{h} = \int_0^1 e^{\mu X(s)} X'(s) e^{(1-\mu)X(s)}\, d\mu = \int_0^1 e^{\lambda X(s)} X'(s) e^{(1-\lambda)X(s)}\, d\lambda.$$

Commuting case. If $X(s)X'(s) = X'(s)X(s)$, then $e^{\lambda X(s)}X'(s) = X'(s)e^{\lambda X(s)}$ for all λ, so:

$$f'(s) = \int_0^1 e^{\lambda X(s)} X'(s) e^{(1-\lambda)X(s)}\, d\lambda = \int_0^1 X'(s) e^{\lambda X(s)} e^{(1-\lambda)X(s)}\, d\lambda = X'(s)e^{X(s)} \int_0^1 d\lambda = X'(s)e^{X(s)}.$$

Similarly, pulling out $e^{X(s)}$ on the right: $f'(s) = e^{X(s)}X'(s)$.

4.4 Commuting Time-Dependent Coefficient Matrices

Theorem 4.4.1: Fundamental Solution for Commuting $A(t)$

Let $n \in \mathbb{N}$, $I \subseteq \mathbb{R}$ open, $A \in C(I, M_n(\mathbb{K}))$ satisfy

$$A(t)A(s) = A(s)A(t), \qquad \forall\, t, s \in I.$$

Fix $t_0 \in I$ and define

$$F(t) = \exp\left(\int_{t_0}^{t} A(s)\, ds\right), \qquad t \in I.$$

Then $F \in \mathcal{F}(I, A)$.

Proof

Define $X(t) = \int_{t_0}^{t} A(s)\, ds$, so $X'(t) = A(t)$. We check the commutativity condition for Exercise 4.3.2:

$$X'(s)X(s) = A(s)\int_{t_0}^{s} A(\mu)\, d\mu = \int_{t_0}^{s} A(s)A(\mu)\, d\mu = \int_{t_0}^{s} A(\mu)A(s)\, d\mu = X(s)X'(s),$$

using $A(s)A(\mu) = A(\mu)A(s)$ for all s, μ. Therefore, by the commuting case in Exercise 4.3.2:

$$F'(t) = (e^{X(t)})' = e^{X(t)}X'(t) = e^{X(t)}A(t) = F(t)A(t).$$

Wait – we need $F'(t) = A(t)F(t)$, not $F(t)A(t)$. Using the left-commutativity: since $X'(t)X(t) = X(t)X'(t)$, the commuting case gives both $e^{X(t)}X'(t) = X'(t)e^{X(t)}$, i.e. $F'(t) = A(t)F(t)$.
To confirm $F \in \mathcal{F}(I, A)$, we also check $\operatorname{rank} F(t_0) = n$: $F(t_0) = e^{X(t_0)} = e^0 = I$, which has rank n. By Lemma 1.2.2, $F \in \mathcal{F}(I, A)$. ■

Chapter A

Summary of Key Formulas

Object	**Formula / Statement**
Solution space	$\dim S(I;A) = n$
General solution	$x(t) = F(t)F(t_0)^{-1}x_0 + \int_{t_0}^{t} F(t)F(s)^{-1}b(s)\,ds$
All fund. solutions	$\mathcal{F}(I,A) = \{FC \mid C \in \mathrm{GL}_n(\mathbb{K})\}$
Liouville	$\det F(t) = \det F(t_0)\exp\bigl(\int_{t_0}^{t} \operatorname{tr} A\,ds\bigr)$
Matrix exponential	$e^A = \sum_{k=0}^{\infty} A^k/k!$
Derivative	$(e^{tA})' = Ae^{tA} = e^{tA}A$
Inverse	$(e^A)^{-1} = e^{-A}$
Determinant	$\det e^A = e^{\operatorname{tr} A}$
Commutativity	$AB = BA \Rightarrow e^{A+B} = e^A e^B$
Duhamel	$x(t) = e^{(t-t_0)A}x_0 + \int_{t_0}^{t} e^{(t-s)A}b(s)\,ds$
Putzer ($\lambda \neq \mu$)	$e^{tA} = e^{\lambda t}I + \frac{e^{\lambda t}-e^{\mu t}}{\lambda-\mu}(A-\lambda I)$
Putzer ($\lambda = \mu$)	$e^{tA} = e^{\lambda t}(I + t(A-\lambda I))$